

MINERAL RESERVE ESTIMATION USING SURPAC SOFTWARE

MI449 - MINE DESIGN PROJECT - I

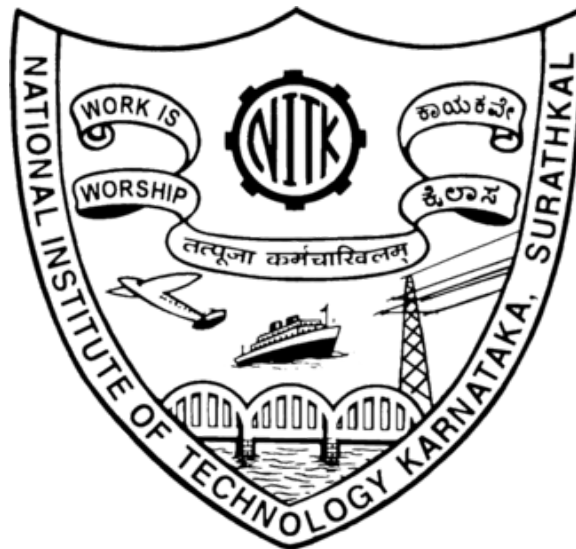
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DEPARTMENT OF MINING ENGINEERING

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Mineral Reserve Estimation Using SURPAC Software

Chapter 1 - Introduction

The magnanimous growth and ever-increasing demand for finished products have pressurized the mother industry (mining) to supply raw materials to meet the increased demand for growth and sustainable development of society. These materials can be extracted from the earth's crust by any of the two methods of mining i.e., surface mining methods and underground mining methods. Surface mining operations can be classified into open pit, open cast, strip, alluvial, and in-situ mining, depending on the method of mining being considered (Hartman, 1987). Open-pit mining is a method of developing and exploiting the deposit by making a void in the surface, also known as a pit, and developing it sequentially to extract the ores. It starts off with a small pit and gradually develops with the lapse of time in a pre-planned manner to take the final shape of the pit called as ultimate pit which includes the initial void by which the deposit was attacked. Opencast mining is the advancement in the open pit mining methods, developed due to the acute shortage of granted leasehold areas where the overburden can be piled for future reclamation purposes. In the open-cast mining method, after all the ores in a particular area have been mined the overburden materials are dumped in-pit in those extracted areas instead of dumping it in the overburden pile within the leasehold area outside the pit. This has proved to be cost-effective and acquires lesser land area and continuous reclamation work is also ensured. At present opencast mining has a larger share of total ore production in India, owing to the technological advancement due to the availability of heavy earth moving machines (HEMMs) which has increased the rate of production while ensuring safety and requires lesser manpower. The increased rate of operation facilitates lesser development time.

The advancement in computer technology has started the development of software applications. SURPAC is one of the software applications that was developed to be used in the mining industry. This SURPAC software should be addressed by a specialized staff composed of mining engineers, surveyors, and geologists. Each of them has a separate module dedicated. The surveyors create a Digital Terrain Model (DTM), geologists create a digital model of the deposit, and mining engineers use it to estimate the production of useful minerals. Underground minerals, metals, and nonmetals are invisible whose shapes, quality compositions, and quantities are not known. Geological explorations and investigations aimed at determining the unknown minerals and metals.

At the start of the process, topographical and lithology data are gathered and a database is generated by using SURPAC. Depth, thickness, and grade changes, ore volume, shape and extensions, and properties are determined by mathematical and algorithm approaches using this database. All numerical estimations are used to get out visuals to bring out the body model. The concrete data to define the shape, location, quality, and quantity of an ore body is by bore cores. GPS data is usually taken to draw topographical maps and surfaces. Underground maps like thickness and grade contours are drawn also. Topographical coordinates are combined with stratigraphic information, a 3-dimensional data set is handled. After following several mathematical techniques, 3- a dimensional ore body model is often obtained. Besides the physical ore model, the quality composition should even be known

This is often crucial because further engineering activities have a cheap aspect to perform. With the known volume of block properties like thickness and grade of ore at each particular block, it becomes possible to convert this information to an economical aspect. (Volume $\times$ tonnage factor $\times$ grade = block reserve.) Surveying includes 3-dimensional components x, y, and z (easting, northing, elevation) which are used for surface modeling. Drill hole data depth and layer information contribute to explaining how the geological structure is in the three dimensions. Drill holes also carry knowledge about ore grade. Geological interpretation of stratigraphic layers provides a 3-dimensional ore body model.

Objective

- To Study the Mineral Resource Estimation of a particular mining project using SURPAC software.
- Develop a model in SURPAC software using Geology, Survey, Borehole data, and Assay Value of a particular deposit.
- Study the Reserve Estimation by Conventional methods and Computer-aided methods and compare them.

Chapter 2 - Literature Review

2.1 Reserve Estimation Techniques:

The samples forming the foundation of all ore reserve estimates constitute a small fraction of the entire deposit under examination for potential mining. Typically, these samples are acquired through diverse methods such as diamond core drilling, reverse circulation drilling, and others. Fundamentally, reserve estimation involves the calculation of volume, followed by the assignment of a value to the grade of that volume. Various mathematical methods have been devised to determine the value to be attributed to the specified volume.

2.2 Conventional Estimation Techniques

Before the introduction of computers and geostatistical methods, mineral resources were assessed and estimated. In the traditional approach, measurements of areas were made, and calculations for volume, tonnage, and average grade were performed using paper, pencil, slide rules, and calculators.

Popoff (1966) outlines the principles and traditional approaches for calculating reserves of mineral deposits, incorporating three main principles:

- (1) the rule of gradual changes
- (2) the rule of nearest points or equal influence
- (3) the rule of generalization.

The rule of gradual change, or the law of linear functions, posits that all sample elements within a mineral body change gradually and continuously as a linear function along a straight line connecting two adjacent sample points. This rule is applicable not only to grade and weight factors but also to other parameters like areas, volumes, and tonnages. The conventional triangles method, used for calculating mineral inventories, employs the principle of gradual changes.

The rule of nearest points or equal sphere of influence suggests that the value of any point between two samples remains constant and is equal to the value of the nearest sample. It assumes that the value of a sample extends halfway to any adjacent sample. The rule of generalization can be described as the empirical method, not truly a rule, often arbitrarily applied based on judgment derived from past experiences and opinions. One common example is assigning a definite weighting factor for reserve computations based on determinations from a similar deposit. In certain instances, the use of the rule of generalization is justified and necessary.

In conventional estimation techniques, ore reserves can be estimated using three different methods: polygonal, rectangular, and/or cross-sectional.

2.3 Basic principle of ore reserve estimation

The basic principle of ore reserve estimation is utilizing the tonnage formula. The unit of estimation is tonnes(t) and the formula is :

$$\text{Tonnage (t)} = \text{Volume (V)} \times \text{Specific Gravity (Bulk density)} \dots\dots\dots(1)$$

The above equation can be compared with the mass volume formula where :

$$\text{Mass} = \text{Volume} \times \text{Density} \dots\dots\dots(2)$$

Volume can be calculated as :

$$\text{Volume (V)} = \text{Area (A)} \times \text{Influence of a third dimension} \dots\dots\dots(3)$$

Area (A) is calculated from the borehole plan or sections.

Influence is determined based on the method of estimation.

2.3.1 Basic Assumptions :

Several assumptions are taken for reserve estimation. They are :

1. The parameters of an ore body established at one point change to an adjoining point in accordance with a certain principle like the rule of gradual changes, the rule of nearest points.
2. The projected continuity of the ore body based on exploration data is supposed by geological setup.
3. The samples are collected with equal precision at every point and they represent the ore zone.

2.4 Polygonal method

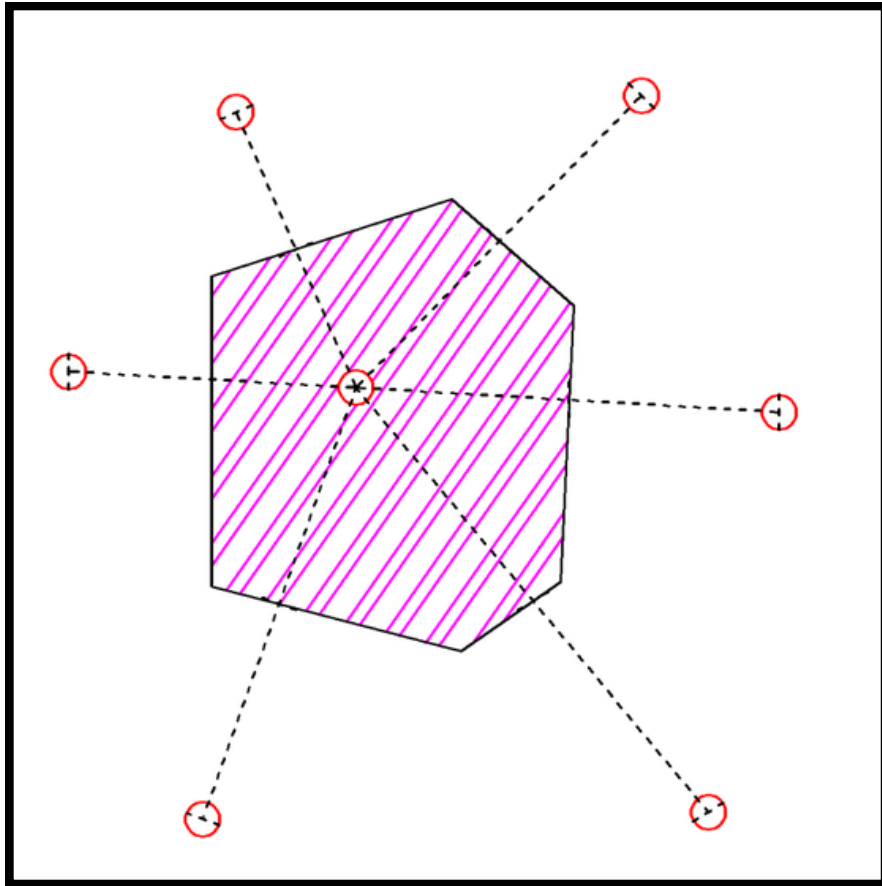


Fig1. Polygonal method of grade estimation

The method is based on an area of influence. The area of influence is determined by constructing polygonal blocks around each hole that extends half the distance between two holes. The average grade and thickness of the hole inside the polygon are assigned to the entire polygon in order to provide a volume for the reserve estimate. The volume is computed by multiplying the area of polygons with thickness. Volume is multiplied with specific gravity to get tonnage. Summation of all polygons will get the tonnage and metal content of the total deposit.

The polygonal method is used for tabular and large lenses or bodies.

Disadvantages of this method are :

1. It gives greater weight to isolated holes.
2. The assay values are used only once.

A variant of the polygonal method uses values of holes at the corners of polygons instead of the center. It will be discussed as a triangular method.

2.5 Triangular Method

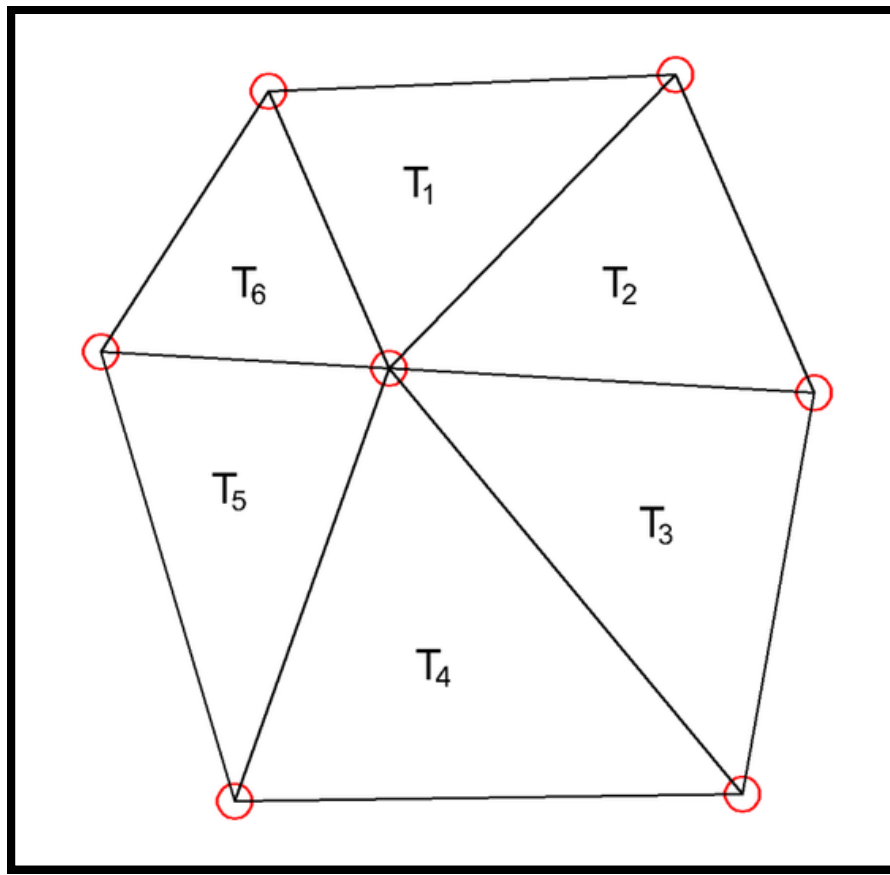


Fig2.Triangular method of grade estimation

This method is a slightly advanced version of the polygonal method. In this method, holes are connected to adjacent holes. This divides the ore body into a series of triangles (T_1 , T_2 , ..., T_n). Each triangle rests on the plane of the map and represents a base area of an imaginary prism with edges t_1 , t_2 , and t_3 equal to vertical thicknesses of intercepted mineralization. This method uses the triangle area between three adjacent holes, the average grade, and the thickness of these holes to compute tonnage.

2.5.1 Computation steps

The following steps are involved in the triangular method-

1. The area of the triangle is calculated using a geometrical formula. Area (A) is multiplied by the average thickness intercepted in holes at three edges of that triangle to get volume (V).
2. Tonnage is computed by multiplying volume (V) with the average specific gravity of the host rock. This will give the tonnage of one triangle.
3. Metal content is estimated by multiplying tonnage (step 2) and average grade.
4. Steps 1 & 2 are repeated for all triangles with positive intercepts at their edges.
5. The total tonnage and metal content of the deposit are arrived at by summation of values calculated for each triangle.

The triangular method is best suitable for flat or gentle dipping ore body having good continuity and correlation.

Different formulas involved in calculation are :

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)} \quad \text{.....(4)}$$

Where s is semi-perimeter of the triangle.

$$s = \frac{(a+b+c)}{2} \quad \text{.....(5)}$$

$$\text{Average thickness} = \frac{t_1(s-a)+t_2(s-b)+t_3(s-c)}{s} \quad \text{.....(6)}$$

$$\text{Average grade} = \frac{t_1g_1(s-a)+t_2g_2(s-b)+t_3g_3(s-c)}{t_1(s-a)+t_2(s-b)+t_3(s-c)} \quad \text{.....(7)}$$

Table1.The datasheet of triangular method will look like below-

Triangle No	Area of Triangle	Avg Grade	Avg. Thickness	Volume	Tonnage	Metal Content
	m ²	%	m	m ³	tonne	tonne
	col. 1	col. 2	col. 3	col. 4 = col. 1 x col. 3	col. 5 = col. 4 x Sp. Gr.	col. 6 = col. 5 x col. 2
T ₁						
T ₂						
-						
-						
-						
T _n						

2.6 Cross Section Method

In this method, ore body is interpreted on cross-section. The ore body is divided into different segments with the help of transverse section lines. The section line can be spaced at equal or unequal intervals based on grid interval and hole locations.

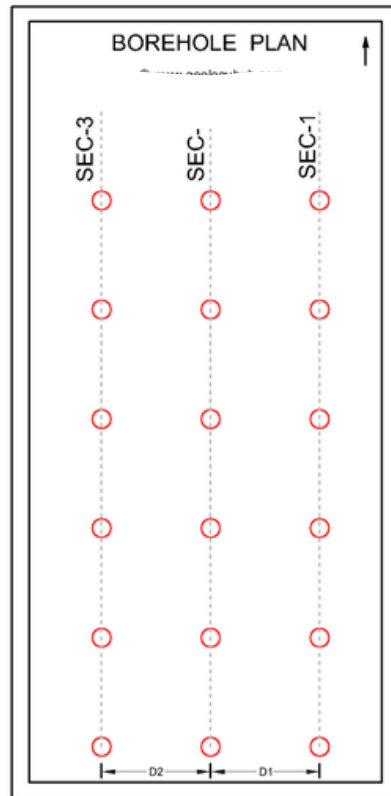


Fig3. Borehole Plan showing holes drilled to explore ore body

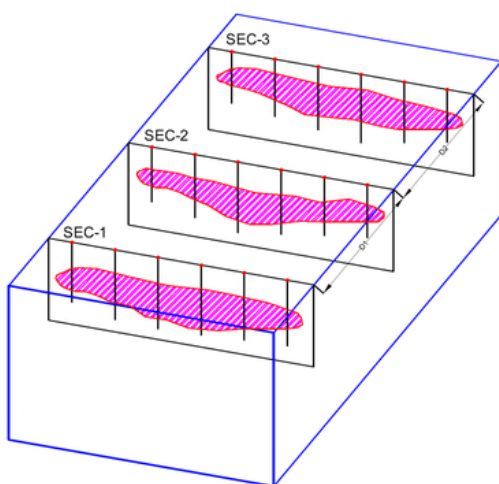


Fig4. Transverse sections showing ore body in third dimension

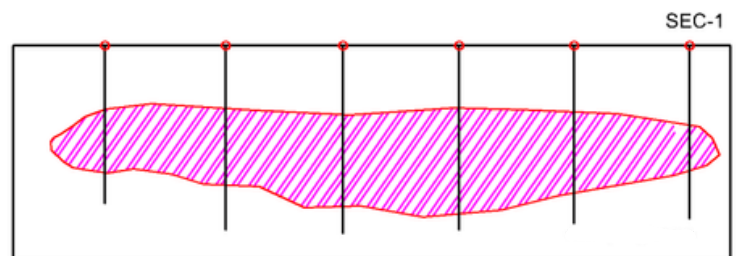


Fig5. Single transverse section utilize to compute sectional area

2.6.1 Computation steps

1. The total ore body is divided into sub-blocks along the section line and a length equal to half of the distance between the adjoining sections.
2. For the computation of reserve, the volume of each sub-block is required. The volume is calculated by the multiplication of the sectional area with half the distance of the adjoining section on each side (i.e. area of influence).
3. The sectional area of the ore body is calculated by a geometric formula. AutoCAD software can be used to measure the area of irregular ore body.
4. Tonnage for each sub-block is calculated by multiplication of volume and specific gravity.
5. The metal content of each sub-block is calculated by multiplying the tonnage and average grade of that sub-block.
6. The total tonnage of the ore body is the summation of sub-block tonnages. Similarly, total metal content is the summation of sub-block metal contents.
7. The average grade of the ore body is the total metal content divided by the total tonnage in percentage terms.

Table2.Datasheet of cross-sectional method will look like below:

Section line	Sectional Area	Avg Grade	Strike Influence ($\frac{D_1+D_2}{2}$)	Volume	Tonnage	Metal Content
	m ²	%	m	m ³	tonne	tonne
	col. 1	col. 2	col. 3	col. 4 = col. 1 x col. 3	col. 5 = col. 4 x Sp. Gr.	col. 6 = col. 5 x col. 2
Sec-1						
Sec-2						
-						
-						
-						
Sec-3						

2.7 Disadvantages of conventional methods of reserves estimation

2.7.1 Limited Spatial Accuracy

Conventional methods often rely on manual measurements and simplistic geometric assumptions, leading to limitations in accurately representing the complex spatial characteristics of mineral deposits.

2.7.2 Subjectivity and Judgment

These methods may involve a significant degree of subjectivity and reliance on the judgment of the geologist or engineer conducting the estimation. This subjectivity can introduce bias and variability in the results.

2.7.3 Time-Consuming

The manual nature of conventional methods, involving tools like paper, pencil, slide rules, and calculators, makes the process time-consuming. This can be a significant drawback, especially when dealing with large datasets or complex deposit geometries.

2.7.4 Inefficiency for Complex Deposits

Conventional methods may struggle to accurately estimate reserves for deposits with irregular shapes, multiple mineralization zones, or intricate geological structures. Complex deposits often require more sophisticated techniques.

2.7.5 Difficulty in Handling Data Variability

These methods might not effectively handle the variability within the dataset, especially when dealing with heterogeneous mineralization. Modern statistical and geostatistical methods are better equipped to account for variability.

2.7.6 Limited Use of Geostatistics

Conventional methods may not fully leverage the power of geostatistical techniques, which can provide more robust and spatially informed estimations. Geostatistics can better capture the spatial correlation and distribution of mineralization.

2.7.7 Insensitive to Local Variations

Conventional methods might not be sensitive enough to local variations within a deposit. This insensitivity can lead to inaccuracies in estimating ore grades and volumes.

2.7.8 Difficulty in Updating Models

Updating models based on new exploration or drilling data can be challenging with conventional methods. It may require significant manual effort and time, leading to delays in incorporating new information.

2.8 COMPUTER AIDED ORE RESERVE ESTIMATION

With the progression of technology and the advent of computers, software developers applied their expertise to link the fundamentals of the estimation process with mathematical functions and equations. This effort resulted in the creation of sophisticated software designed for estimating ore reserves with increased speed and enhanced accuracy. Interestingly, these software applications still incorporate the three fundamental principles of conventional methods outlined by Popoff (1966).

Additionally, geostatistical tools emerged as valuable resources for ore reserve estimation through computer applications. The three commonly employed techniques in computer-assisted ore reserve estimation include nearest neighbor, inverse distance, and kriging.

2.8.1 Surpac:

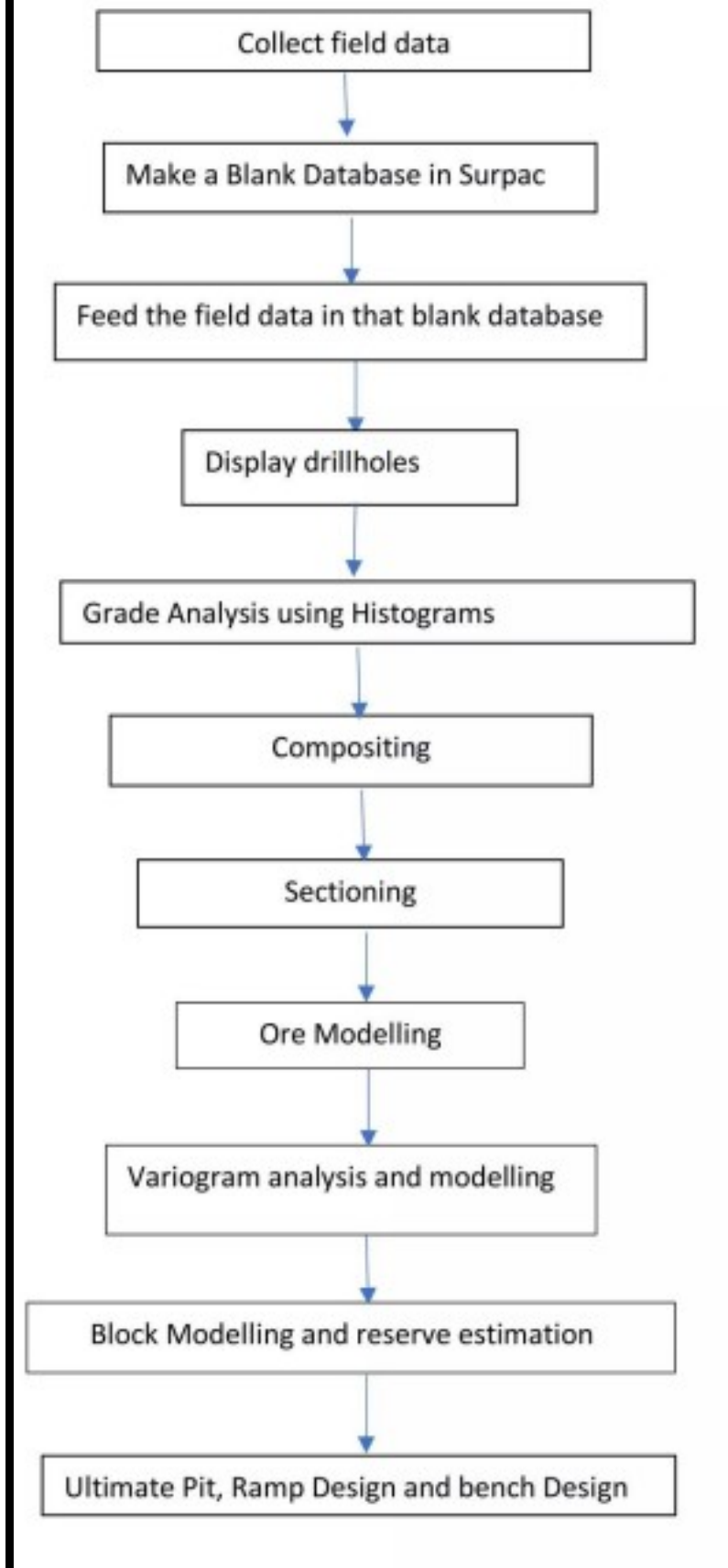
In this project, we have used GEOVIA Surpac for studying and further planning of the mine using the data provided by the instructor.

GEOVIA Surpac is the world's most popular geology and mine planning software, supporting open pit and underground operations and exploration projects.

Using Surpac, we can visualize an interpolation, lock it in 3-D, and then we can bring in all the drilling, all the infrastructure, the development, the headings, and know instantly whether we are on track.

It also allows us to integrate various processes to result in a well-designed plan.

Overall Workflow



Chapter 3 - Methodology

3.1. CREATION OF A GEOLOGICAL DATABASE

For the process of geological modeling, the first step is to create a geological database. The geological database is basically done from the raw data acquired from the boreholes or equivalently drill holes so that the software understands the type of information fed into its system and can compute the results or display any graphical form according to the user's requirements. For feeding into a Surpac database or basically any geological database two mandatory tables are required - collar and survey. For obtaining information related to the grade, types of rocks present near the orebody, and the presence of any geological features or discontinuities, other optional tables are used such as assay, litho, geology, etc. The software system uses two tables intrinsically as a part of the system tables - translation and styles which deal with the graphical display formats.

During the process of data importing, data is taken from the CSV files (the given data files) to the newly created Surpac database through the software wizard. Data can be imported from access and various versions of Paradox database files. It has to be made sure that the correct CSV file should be selected for the tables and the column number of Surpac database parameters and the field number of the parameters mentioned in CSV files should match each other.

The starting point of all mining projects is the drill-hole data. It continues the basis on which feasibility studies and ore reserve estimation are done. A number of tables are included in a geological database, containing different data. Each table contains a number of fields, having many records, and with each record containing the data fields. Surpac uses a relational database model and supports several different types of databases, i.e. oracle, paradox, and Microsoft Access.

Two mandatory tables are required in Surpac within a database:

1. Collar
2. Survey.

3.1.1 Collar table:

The information in this table describes the location of the drill-hole collar, the maximum depth of the hole, and whether to calculate a linear or curved hole trace when retrieving the hole. For each drill hole, optional collar data may also be stored. For instance, data drilled, type or drill-hole, or project name.

In a collar table, the mandatory fields are as given below:

1. Hole_id
2. Y
3. X
4. Z
5. Max_depth
6. Hole_path

3.1.2 Survey table:

It stores the drill-hole survey information, used for calculating the drill-hole terrace co-ordinates.

The mandatory field in a survey table includes:

1. Down-hole survey depth,
2. Dip, and
3. Azimuth of the hole.

Other Optional Tables:

Apart from the mandatory tables, the optional tables include geology and assay. These tables are added and used to store information.

Importing Data:

While importing data two things are important to pay attention to, such as

- 1) Naming the format file
- 2) Checking the “overlapping sample check”

The name of the format file should be the same as the database name for simplicity as both files are required for use in other systems.

Overlapping of data may occur between the assay and lithology data as they refer to the same interval of a borehole. The process of lithology and assay sampling is elaborated in Fig. 2. Therefore checking of the overlapping should be done.

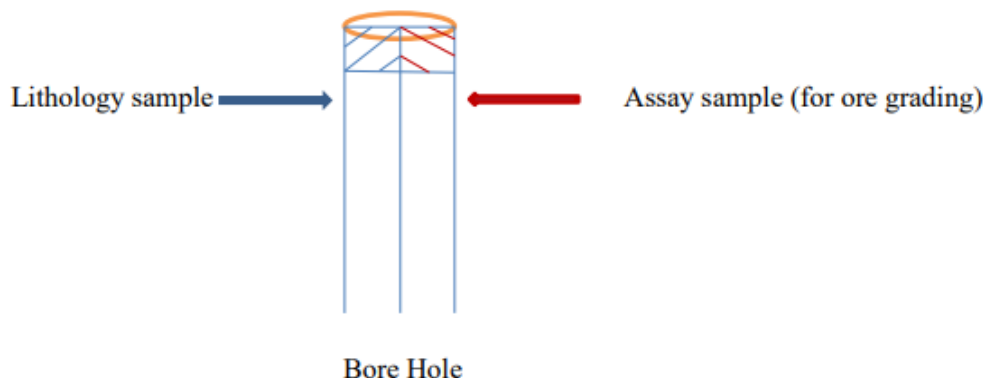


Fig7. Lithology and assay sampling

3.3. 3-D MODELLING

Prior to 3-D modeling, some of the other functions to be carried out include the generation of surface DTM from a given borehole set, indicating the depths in the boreholes which can be considered as ore, and the creation of sections for different drill hole sets. The surface string is generated using the 'import data from one file' function.

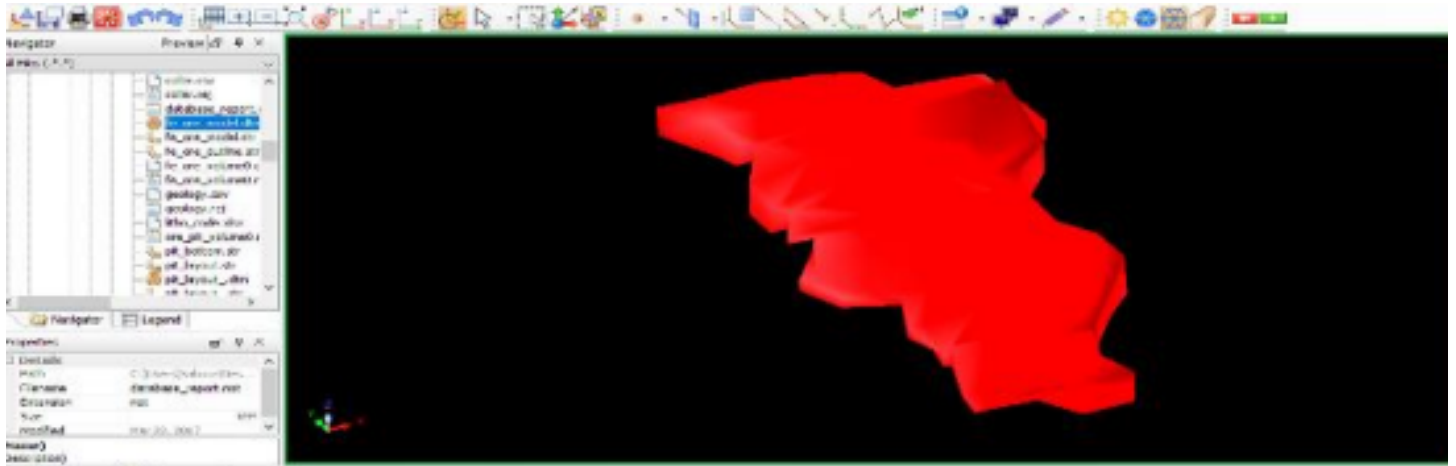


Fig9.3-D ore body model

A 3D model of the ore body is the best resource for visualizing the characteristics of the deposit. The ore body was developed by triangulating the segments of the mineralization zone which were sectioned and digitized from the boreholes. The segments that are to be used for developing the 3D model of the orebody should be first cleaned. The cleaning process should be fulfilled on the string of the segments. Out of all the tests, 3 tests (i.e. cross-overs, duplicated points, and spikes) could fulfill the requirements. After completing of the tests, the segments are joined together as per the requirements and behavior of the segments. Then, the newly formed 3D model is validated, if the validation is successful the volume and the area of the orebody can then be obtained by producing the “report volume of solids”.

3.4. BLOCK MODELLING

Before starting block modeling, the compositing of drill holes is done for proper resource evaluation. In the process of compositing, the assay values of the adjacent samples from the drill holes are combined in order to determine the assay values of longer drill hole intervals. Basically, there are six types of compositing methods used in Surpac software- downhole, bench elevation, grade constraints, geological constraints, from the end of the hole, and from multiple elements

Next comes the process of block modelling in which the solid ore model is divided into smaller regular block units of customized size for better resource evaluation which means determining the volume of the entire ore deposit from the block model and second, to estimate the grade of a particular block using different grade estimation techniques with the aid of compositing.

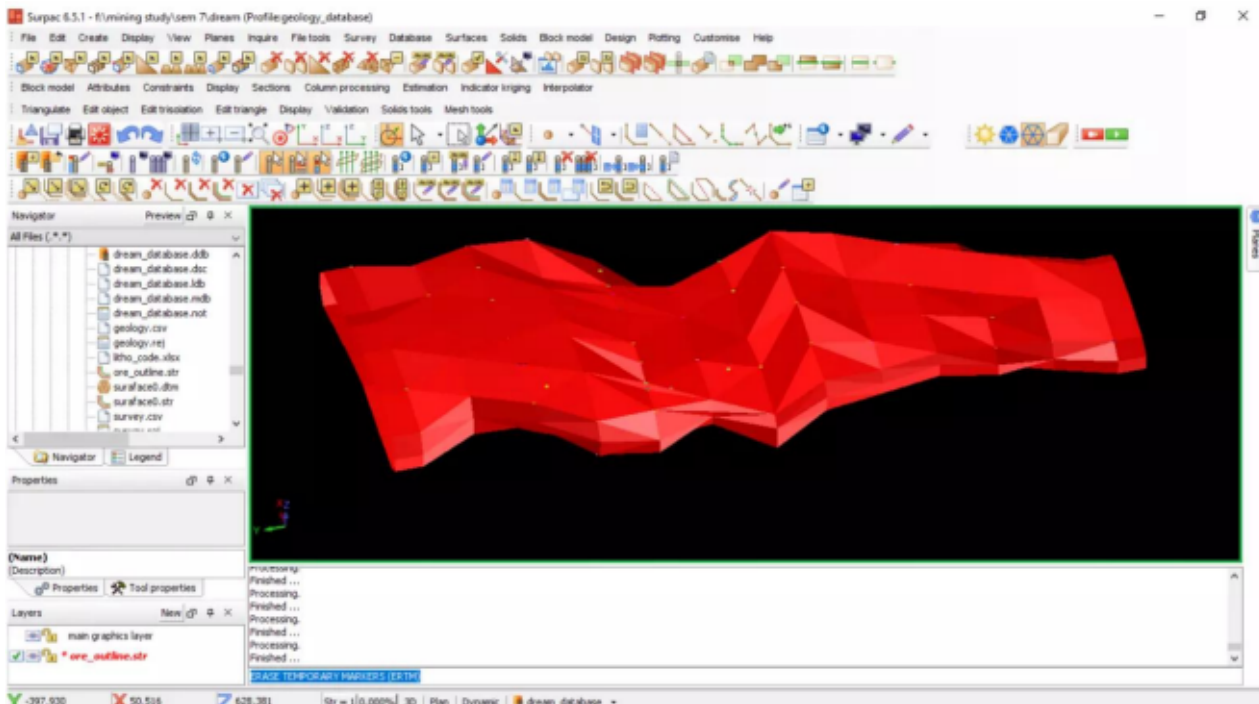
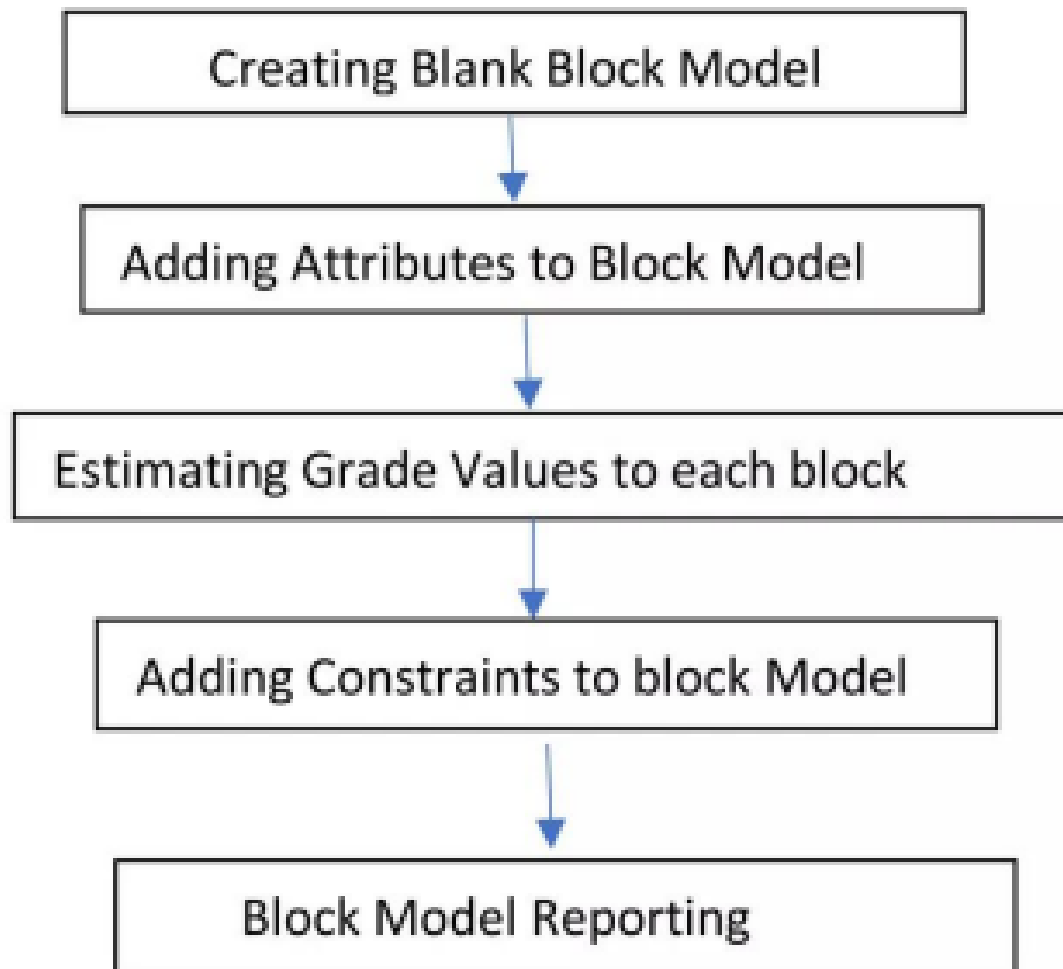


Fig10.Block model

Work Flow



4. Conclusion

The mineral reserves estimation project utilizing SURPAC software has proven to be an invaluable asset in the mining industry, significantly enhancing the accuracy, efficiency, and reliability of reserves assessments. The importance of reserves estimation cannot be overstated, as it plays a pivotal role in various stages of mining operations, from initial exploration to mine planning and resource optimization.

Reserves estimation serves as the foundation for informed decision-making throughout the mining life cycle. During exploration, it provides a basis for assessing the economic viability of a potential site, guiding investment decisions and risk management.

The comparison between conventional methods and computer-aided techniques like SURPAC underscores the transformative impact of technology on the mining industry. Traditional approaches often relied on manual calculations and simplified models, leading to potential inaccuracies and inefficiencies. In contrast, SURPAC's advanced algorithms, 3D modeling capabilities, and data integration features enable more precise and detailed reserve estimations. The software facilitates the incorporation of diverse geological and geospatial data, allowing for a comprehensive and dynamic analysis of mineral deposits.

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